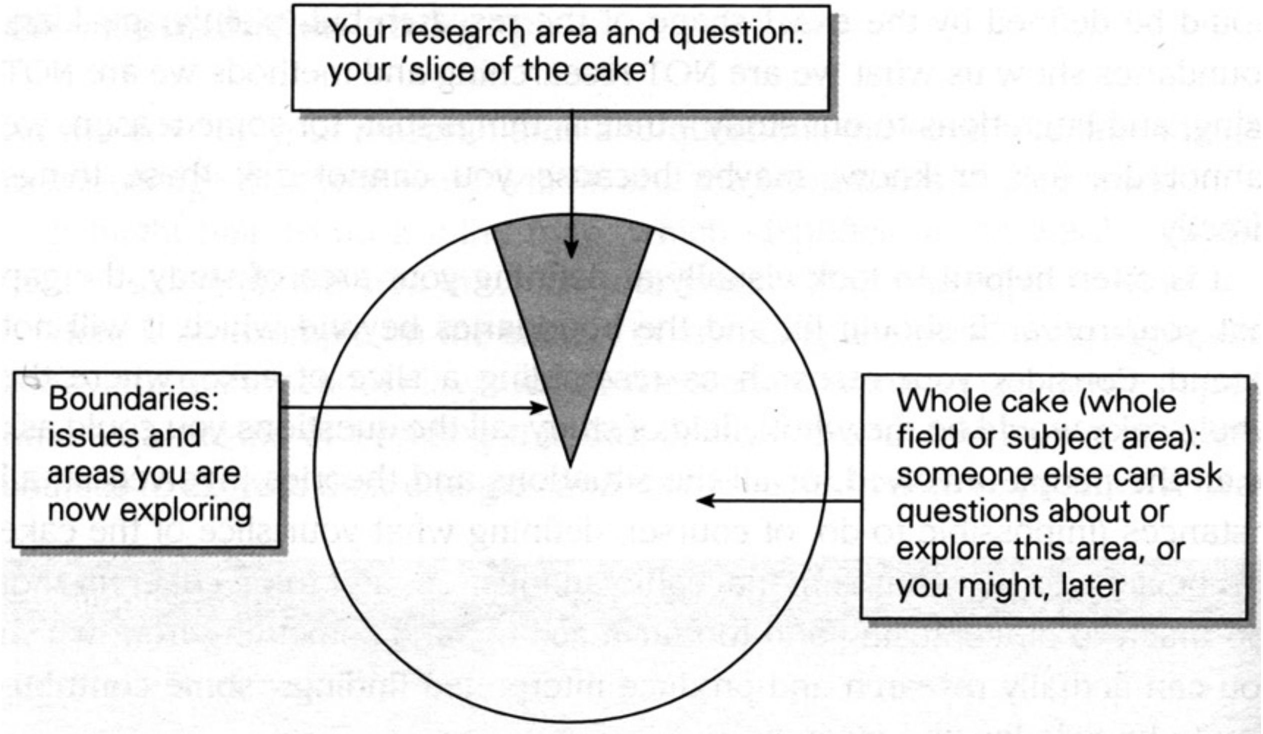
RESEARCH METHODS: QUANTITATIVE

COMM50 Postgraduate Study & Research Methods

Formulating Research

- Remember, research will often follow from knowledge of a subject domain
  - …and a literature review that reveals a *niche*

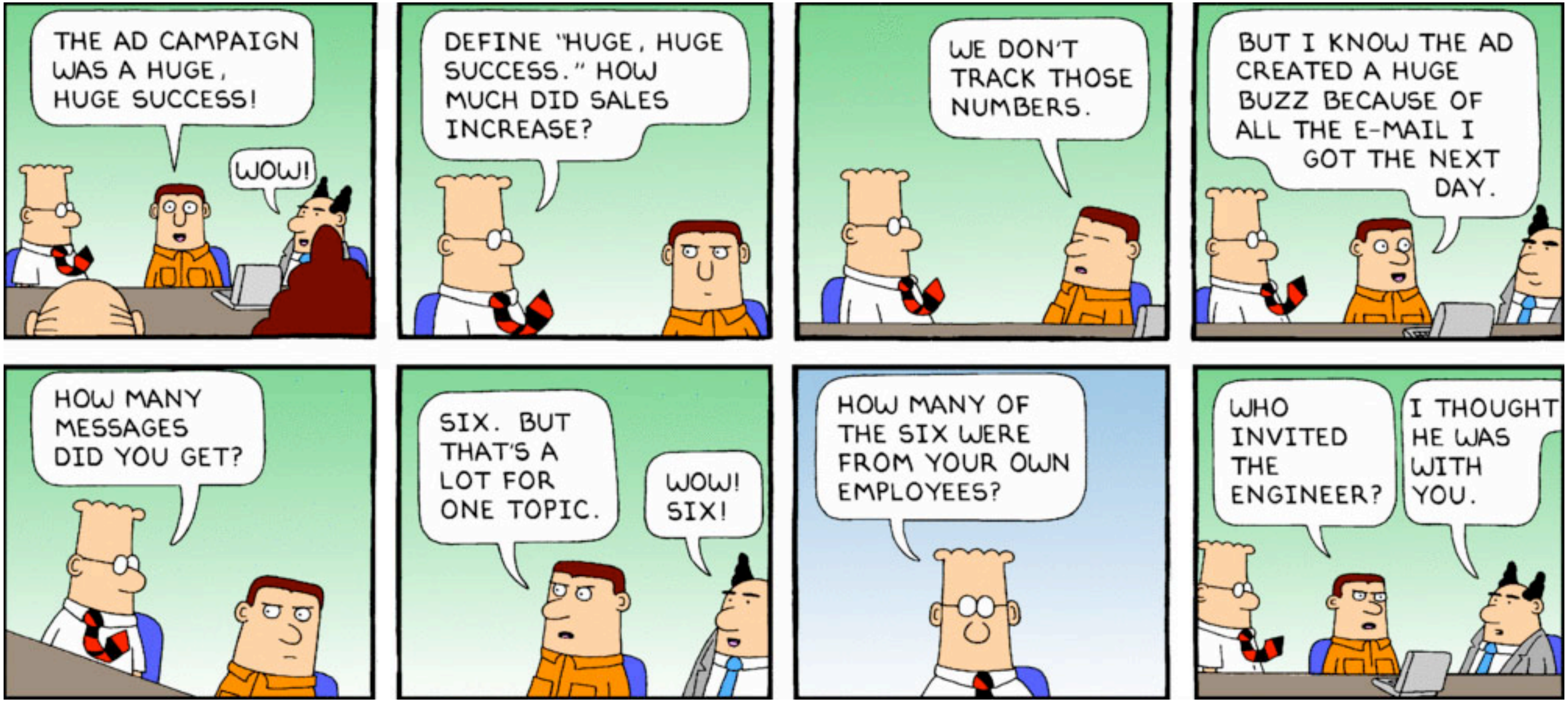


* *Wisker, G. (2008). The Postgraduate Research Handbook, 2nd Edition, Palgrave, London, UK.*

Quantitative & Qualitative

- Quantitative
  - Is deductive and linear
  - Begins with a testable hypothesis
  - Generally reliant upon collecting *objective* data
- Qualitative
  - Is iterative and inductive
  - Begins with a broad question or investigation area
  - Usually explores *subjective* data

Quantitative & Qualitative



Qualitative Versus Quantitative Research

Comparison Dimension	Qualitative Research	Quantitative Research
Types of questions	Probing	Nonprobing
Sample size	Small	Large
Information per respondent	Much	Varies
Administration	Requires interviewer with special skills	Fewer special skills required
Type of analysis	Subjective, Interpretive	Statistical, summarization
Hardware	Tape recorders, projection devices, videos, pictures, discussion guides	Questionnaires, computers, printouts
Ability to replicate	Low	High
Training of the researcher	Psychology, sociology, social psychology, consumer behavior, marketing, marketing research	Statistics, decision models, decision support systems, computer programming, marketing, marketing research
Type of research	Exploratory	Descriptive or causal

** University of Connecticut , 2009*

	Quantitative	Qualitative
Purpose	To determine cause and effect relationships.	To describe ongoing processes/ phenomena in the real world.
Hypotheses	These are stated before the study and tested during the study.	These are usually developed and refined during the investigation. Questions raised determine where the study goes.
Theories	These deductively determine the study.	These are developed at the end of the study by inductive reasoning.
Variables	These are controlled and manipulated to shed light on the hypothesis.	No explicit variables. The focus of qualitative research is to study naturally occurring phenomena to discover important variables.
Data collection	It is important that data are collected objectively.	Data can involve subjectivity (e.g. interaction with participants and the aim is to build rapport (i.e. through interaction with participants).
Design of study	This is stated at the outset and does not change.	This can change as the study develops. The design is flexible.

* *Clarke, D. (2003). Research Methods in Education, unpublished manuscript, Melbourne. In: Davies, M. (2011). Study Skills for International Postgraduates, Palgrave, London, UK.*

	Quantitative	Qualitative
Presentation of Data	This is usually numerical (e.g. statistical).	This is usually done in the form of analysis using language (narrative form) or pictorally (e.g. through the use of video, photographs, drawings, etc.).
Validity and reliability	This is ensured by means of statistical tests.	This is ensured by means of triangulated (multiple sources) of data/evidence.
Data samples	These are carefully chosen to represent larger populations.	Individual cases are studied to shed light on other groups/cases.
Threats to validity	Avoided by means of statistical methods.	Avoided by means of logical analysis to rule out alterative explanations.
Subject of analysis	Simplified and reduced as much as possible.	Studied as a whole, as phenomena occur within reality as a complex system.
Conclusions	These are stated with statistical measures of confidence (e.g. alpha levels).	These are suggestive and always tentative and subsequent studies can cast doubt on them.

** Clarke, D. (2003). Research Methods in Education, unpublished manuscript, Melbourne. In: Davies, M. (2011). Study Skills for International Postgraduates, Palgrave, London, UK.*

Quantitative & Qualitative

- Quantitative
  - Concerned with information that is/can be expressed numerically
  - Allowing the use of graphs, statistics, functions, etc. for analysis & presentation
- Qualitative
  - Concerned with information that is/can be descriptive
  - Allows the use of language analysis, transcripts, summaries, etc. for analysis & presentation

Discussion!

Can you describe a research investigation that uses:

- *A quantitative approach?*
- *A qualitative approach?*

Definitions of Research

Research is a systematic enquiry which is reported in a form which allows the research methods and the outcomes to be accessible to others

-- Allison *et al*. (1996)

Definitions of Research

A systematic process of inquiry consisting of three elements or components:

(a) a question, problem or hypothesis;

(b) data; and

(c) analysis and interpretation of data.

-- Nunan (1992)

Why do we need Quantitative?

Back to Facts…

- The general perception of scientific work is often that we search for fact
- However, researchers generally only exert a belief in something and are often seeking for it to be revised or even debunked
- This comes down to evolution of tools and techniques, expanding knowledge, an the need to account for bias and subjectivity
- Being constrained by prior belief also forces current thinking into silos and impede progress

…and back to Ignorance

- Ignorance being:
  - *"…the absence of fact, understanding, insight, or clarity about something…a communal gap in knowledge"* (Firestein, 2012)
  - *"I know one thing: that I know nothing"* (Socrates) – according to Plato – maybe…
- Does ignorance precede knowledge or does it follow knowledge?
  - Ignorance: questions
  - Knowledge: answers

Quantitative View

- Assumes that the researcher can be separate from the subject
- Relies upon belief that there is a single reality that exists (largely a Platonic view)
- Aim to collect objective data about a phenomena, by exerting a form of control
- May or may not be empirical, but for this purposes of this lecture, we will focus upon the empirical
  - Evidence is collected, often in the form of data

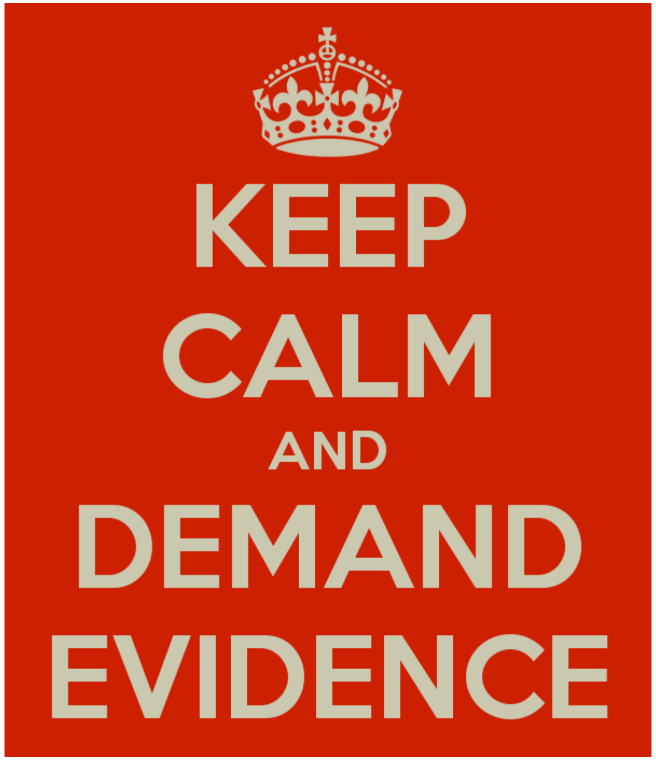
Quantitative Methods

- Focuses on *quantity*
- Research begins with a hypothesis
- Testing
- Controls, measures, examines relationships between variables
- Observation of behaviour or outcomes
- Explanatory: aims to account for cause and effect

Quantitative Methods

- Setting is controlled
- Research sets out to elicit facts and figures
- Methods include:
  - Surveys & questionnaires
  - Experiments & randomised controlled trials
- Data collection is based on measurement
- Data collected in numbers (hard data)
- Undertaking statistical tests on data
- Facts are found to fit the theory

What Constitutes Evidence?



Collection and analysis of data collected that indicates a result, in which we have confidence beyond reasonable doubt.

Aims of Research

- To enhance human rationality and stripping it of myths and superstitions by searching for *truth*
- To achieve maximum information
- To strive for simplicity
- To seek after valid knowledge
- The findings of research should have utility

Quantitative Research Purposes

- Different purposes of research:
  - Description (fact finding)
  - Exploration (looking for patterns)
  - Analysis (explaining why or how)
  - Prediction (forecasting the likelihood of particular events)
  - Problem Solving (improvement of current practice)

Discussion!

Can you describe a situation where each of these types of research is applied:

- *Description?*
- *Exploration?*
- *Analysis?*
- *Prediction?*
- *Problem Solving?*

Hallmarks of Scientific Research

- Purposiveness
  - There must be an *aim* that is testable
- Rigour
  - Sound *methodology* and *context* of the work
- Isolating variables
  - What will be *observed* and will be *manipulated*?
- The hypothesis
  - What is being *tested* and what is the focus?
- Testability
  - Ensuring the test will *answer* the question

Hallmarks of Scientific Research

- Replicability
  - Others able to *repeat* the work. Needs clear methodology
- Precision & Confidence
  - Demonstrating the *accuracy* of the work
- Objectivity
  - Conclusions drawn from the data and *represent fact*
- Generalisability
  - Can your findings put into a *wider* context? (not always)
- Parsimony
  - Small, *meaningful* research – don’t try to do too much

Discussion!

As a researcher, what can you do to increase your chances of meeting each of the 10 hallmarks of scientific research?

Scientific Research Process

1. Identifying a problem
2. Reviewing existing knowledge
3. Defining a population
4. Sampling
5. Data collection design and implementation
6. Data processing and analysis
7. Writing a research report

Research Methods

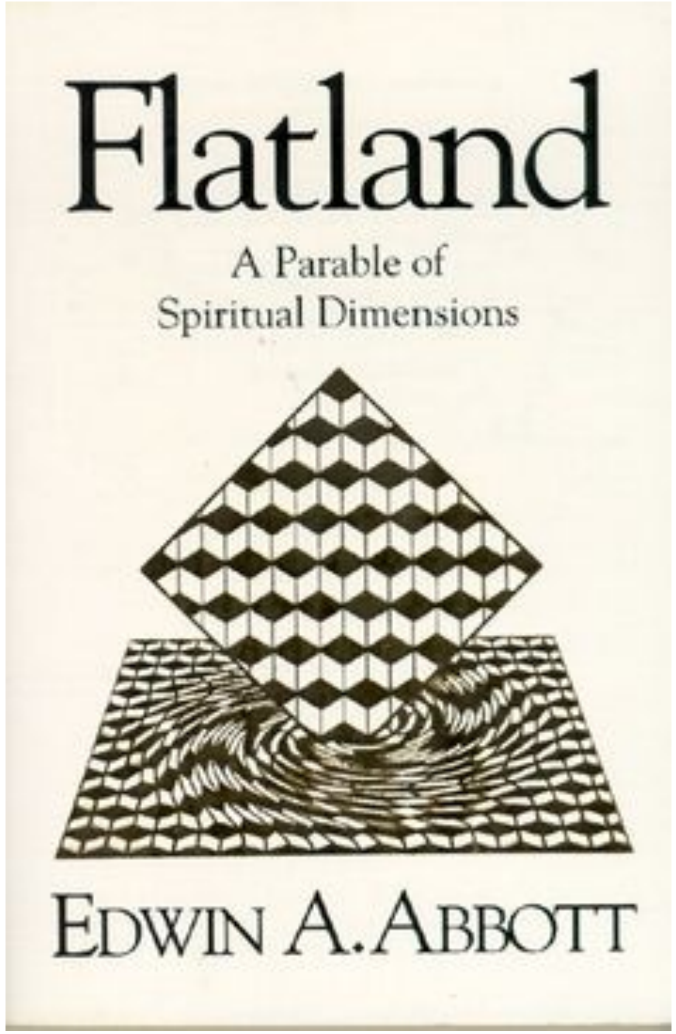
- [Good] scientific research is not a random process
  - Not as haphazard as trial-and-error
  - Imagine the consequences!
- A structure or framework is needed
  - Within which a research plan is formed
  - Enter the methodology

Experimental Process

- State hypothesis based on "causal" relationship between modes/configurations
- Selection of independent variable(s) (the *cause*) and a dependent variable(s) (the *effect*) in relationship
- Design of a controlled experiment or survey
- Data collection
- Data analysis using statistical methods
- Provide evidence / rationale that proves or disproves the hypothesis

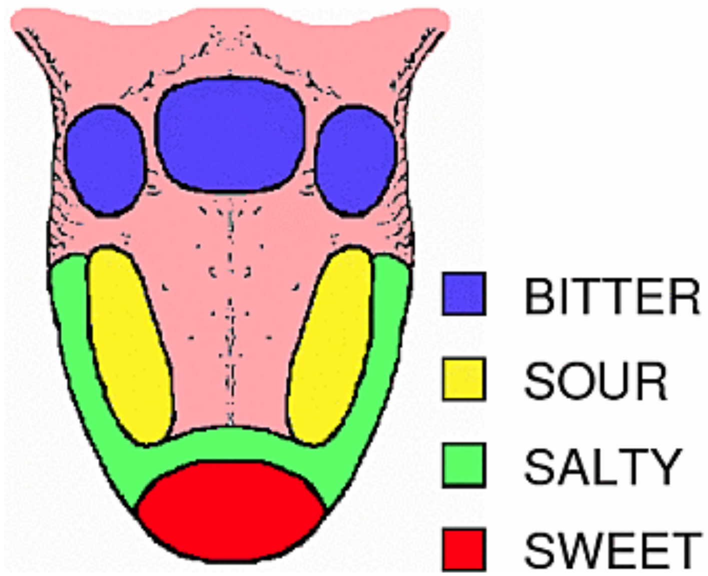
Positivism

- Dominant scientific philosophy up until the early 20th Century
- Seeks a distinct truth that exists and can be experienced and observed, rooted in logic
- Crucially seeking ‘positive’ evidence to substantiate a claim, then becoming fact
- Rooted in logic



Getting It Wrong!

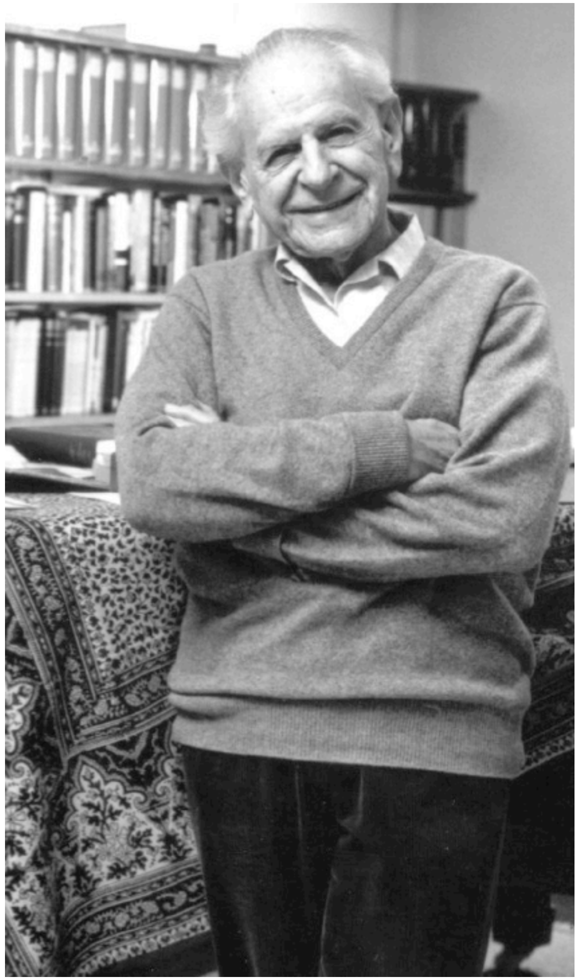
- How does your sense of taste work?



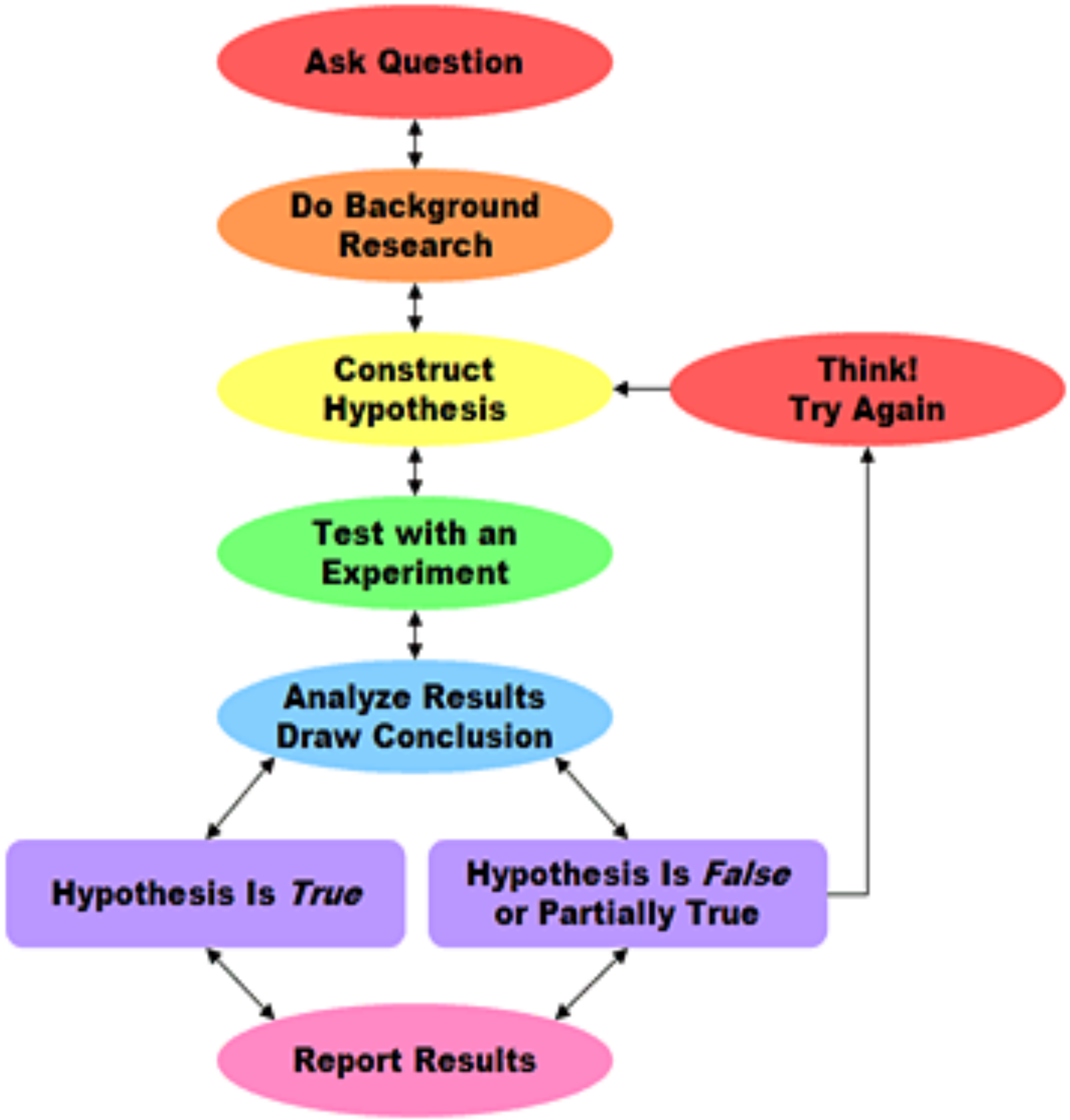
- A mistranslation of DP Hanig's work (in German) by Edward G Boring

Post-Positivism

- Places greater emphasis on the ability to find evidence against a hypothesis
  - *Falsifiability; seeking to refute a hypothesis*
- Maintaining belief in a phenomenon but being open to criticism and evidence that may disprove it
- Critical realism / 'moderated cynicism'



Scientific Method



Hypotheses

- A good hypothesis is testable
  - Not provable, in the sense of "shown to be true"
  - This is standard scientific practice
- Simply refuting a hypothesis is OK but better science will explain why a hypothesis is wrong, and (better still) offer an alternative hypothesis
- Remember, we want defined outcomes, not surrogate or cherry-picked ones!

Hypotheses

- Based on a post-positivist approach, we generally consider the following:
  - Null Hypothesis (H_0): the 'default' position that whatever phenomenon we are interested in is not significantly influenced by another variable or object (any change we see is a result of chance)
  - Alternative Hypothesis (H_1): that there is some causal relationship between the phenomenon and some other variable or object
- We seek evidence indicating we should reject the Null Hypothesis in favour of the Alternative Hypothesis

Hypotheses: Examples

- Null Hypothesis: Changing network protocol at peak hours will not result in reduced latency for individual end users (something other than protocol is responsible for the latency)

- Alternative Hypothesis (the idea or intervention): Changing network protocol at peak hours will result in reduced latency for individual end users

Scientific Method and Swans

Observation:	Every swan I have ever seen is white.
Hypothesis:	All swans are white.
Test:	A random sample of swans from each continent where swans are indigenous identifies only white swans.
Conclusion:	The research indicates that swans are always white, wherever they are observed.
Verification:	Every swan any other scientist has ever observed in any country has always been white.
Theory:	All swans are white.
Prediction:	The next swan I see will be white.

Dahlberg and McCaig (2010) p.20

Designing an Experiment

- What are you trying to find out?
- What information or indicators will be useful?
- Who is your research about (the population)?
- Who are, and how will you access, your participants (the sample) and distribute the survey?
- How much time (and other resources) do you have? Do you have any incentives?
- Are there any ethical issues?

Discussion!

Where have you encountered scientific method before?

What are the limitations of a scientific/quantitative approach?

Scientific Research: Spot the Mistakes

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